



A Project Report on

**Spectrogram-Based Infant Emotional State
Recognition Using Deep CNN**

Submitted by

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July 2026

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Abstract

Infant crying is the primary means of communication through which newborns express their needs such as hunger, discomfort, pain, or fatigue. Identifying the emotional state of infants through their crying sounds can help caregivers respond quickly and appropriately. In this project, a spectrogram based infant cry classification system is developed using Deep Convolutional Neural Networks (Deep CNN). The infant cry audio signals are first preprocessed and converted into spectrogram images to represent the time–frequency characteristics of the sound signals. These spectrogram images are then used as input to a Deep CNN model to automatically learn important features and classify different infant cry emotions. The dataset is divided into training, validation, and testing sets to train and evaluate the model. This approach demonstrates the potential of deep learning techniques for automatic infant cry emotion recognition and can assist caregivers in better understanding infant needs.

Keywords: Infant Cry Classification; Spectrogram images; Deep Convolutional Neural Network; Audio Signal Processing; Emotion Recognition; Deep Learning.

1 Introduction

Infant crying is the primary form of communication used by newborn babies to express their needs and discomfort [1]. However, accurately interpreting different types of cries such as hunger, pain, sleepiness, or discomfort can be challenging for caregivers and medical professionals [1]. With the advancement of machine learning and signal processing techniques, it has become possible to analyze audio signals and automatically identify patterns in infant crying behavior [1]. This project focuses on developing an intelligent system that can classify infant cry sounds using deep learning techniques.

The significance of this research lies in its potential to assist parents, caregivers, and healthcare providers in better understanding infant needs and responding more quickly to their conditions. Early and accurate detection of infant requirements can improve caregiving efficiency and reduce stress for both babies and caregivers [1]. By combining audio signal processing with artificial intelligence, this study contributes to the development of smart healthcare monitoring systems [1].

The main objective of this project is to design and implement a deep learning-based model capable of classifying infant crying sounds according to their underlying causes. In this work, infant cry audio signals are first converted into spectrogram images using the Short-Time Fourier Transform (STFT), which captures the time-frequency characteristics of the audio signal [1]. These spectrogram representations are then used as inputs to a Deep Convolutional Neural Network (CNN) that automatically extracts relevant features and performs classification of different crying patterns.

The scope of this work includes dataset preparation, audio preprocessing, spectrogram generation, deep feature extraction using CNN, and performance evaluation of the classification model. The proposed system aims to achieve reliable and accurate cry classification that can support intelligent infant monitoring applications.

The remainder of this report is organized as follows. The literature review section discusses existing research related to infant cry analysis and machine learning approaches. The methodology section explains the dataset collection, feature extraction process, and CNN architecture used for classification. The results section presents the experimental evaluation and performance metrics of the proposed model. Finally, the conclusion summarizes the outcomes of the study and suggests possible directions for future improvements.

2 Literature Review

This chapter presents an in-depth analysis of existing research work related to infant cry classification. It reviews various approaches previously proposed by researchers, including traditional signal processing methods and advanced machine learning and deep learning techniques. The chapter explores feature extraction techniques such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and spectral features, which have been widely used in earlier studies. It also discusses the application of classification algorithms like Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees, and Artificial Neural Networks (ANN).

In addition, recent advancements using deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are analyzed in detail. A comparative study of these methods is provided, highlighting their strengths, limitations, and performance metrics. The chapter further identifies research gaps, such as limited dataset availability, lack of generalization, and challenges in noise handling, which justify the need for the proposed system. This critical analysis forms the foundation for selecting an appropriate methodology in the subsequent chapters.

3 Objectives

The main aim of this research work is to develop an automated system for identifying and classifying infant crying sounds using deep learning techniques. The proposed system utilizes audio signal processing and Convolutional Neural Networks (CNN) to analyze the time–frequency characteristics of infant cries and accurately determine different crying patterns. The objectives of this work are set out below:

- To develop a spectrogram-based infant cry classification system using Deep CNN to automatically identify different infant emotional states.
- To evaluate the performance of the model using training, validation, and testing datasets for accurate infant cry emotion recognition.

4 Methodology

4.1 Block Diagram

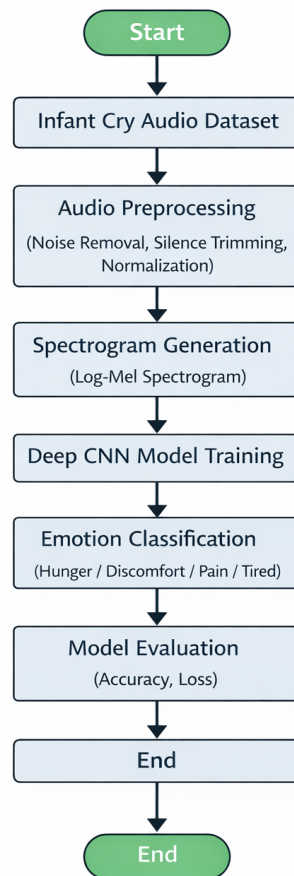


Figure 1: Block Diagram

4.2 Working Principle

The proposed work focuses on developing an intelligent infant cry classification system using deep learning techniques to automatically identify the emotional state of infants. Infant crying is the primary mode of communication, and analyzing cry signals can help in understanding different needs such as hunger, pain, discomfort, and tiredness [1]. Traditional approaches rely on manual feature extraction, which may not effectively capture the complex characteristics of cry signals. Therefore, this work aims to utilize deep learning-based methods for automatic feature learning and classification.

The first step in the proposed system involves collecting infant cry audio data from publicly available datasets such as Kaggle. These datasets contain labeled cry samples corresponding to different infant conditions. After data collection, preprocessing is performed to improve the quality of the audio signals. The cry signals are then converted into spectrogram images using the Short-Time Fourier Transform (STFT), which provides a time–frequency representation of the audio signal [1]. This transformation helps in capturing important patterns present in the cry signals.

The generated spectrogram images are used as input to a Deep Convolutional Neural Network (Deep CNN) for feature extraction and classification. The model automatically learns relevant features from the spectrogram images without the need for manual feature engineering [1,5]. The dataset is divided into training, validation, and testing sets to ensure proper model evaluation. During training, the model learns patterns from the data, while validation helps in tuning the model parameters and preventing overfitting.

The performance of the model is evaluated using accuracy and loss metrics obtained from training and validation processes. Graphs such as accuracy curves and loss curves are analyzed to understand the learning behavior of the model. These results help in assessing how well the model is able to generalize to unseen data. Proper evaluation ensures that the model performs reliably under different conditions.

Finally, a prediction module is developed using a `predict.py` script, where new infant cry audio inputs are processed and classified into corresponding emotional categories. The input audio is converted into a spectrogram, and the trained model predicts the emotion of the infant cry. This enables real-time or offline prediction of infant needs. The overall system provides an efficient and automated solution for infant cry analysis, making it useful for healthcare and smart monitoring applications.

4.3 Observations

To evaluate the performance of the proposed infant cry classification system, several standard performance metrics are used. These metrics help in analyzing the classification accuracy and effectiveness of the model. The commonly used evaluation metrics include accuracy, precision, recall, F1-score, and confusion matrix. These metrics provide a comprehensive understanding of the model performance.

Accuracy: Accuracy is defined as the ratio of correctly predicted samples to the total number of samples. It gives an overall measure of the model performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

Precision: Precision is defined as the ratio of correctly predicted positive samples to the total number of predicted positive samples. It indicates how accurate the model is in predicting positive instances.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall: Recall, also known as sensitivity, measures the ability of the model to correctly identify all positive samples. It reflects how well the model captures actual positive cases.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

F1-Score: F1-score is the harmonic mean of precision and recall. It provides a balance between precision and recall, especially when there is an uneven class distribution.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

Confusion Matrix: A confusion matrix is used to visualize the performance of the classification model. It shows the number of correct and incorrect predictions made by the model. The matrix consists of four components: true positives, true negatives, false positives, and false negatives. It helps in understanding the classification performance for each class.

Training and Validation Metrics: In addition to the above metrics, training accuracy, validation accuracy, and loss curves are analyzed to evaluate the learning behavior of the model. These metrics help in understanding whether the model is overfitting or underfitting during training. A proper balance between training and validation performance indicates a well-trained model.

5 Results and Discussion

5.1 Spectrogram Representation

Spectrograms for Dataset 1: The audio signals from Dataset 1 are preprocessed and converted into spectrogram images using the Short-Time Fourier Transform (STFT). This dataset includes cry categories such as belly pain, hunger, tired, scared, discomfort.

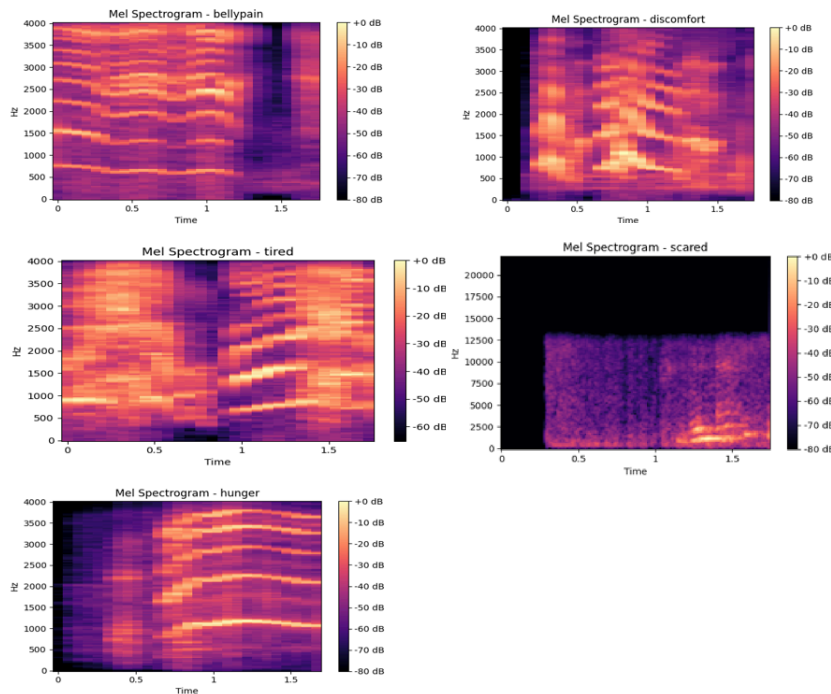


Figure 2: Spectrogram Images of Dataset1

Spectrograms for Dataset 2: Similarly, the audio signals from Dataset 2 are converted into spectrogram images after preprocessing. This dataset includes cry categories such as belly pain, hunger, discomfort, tired, and scared.

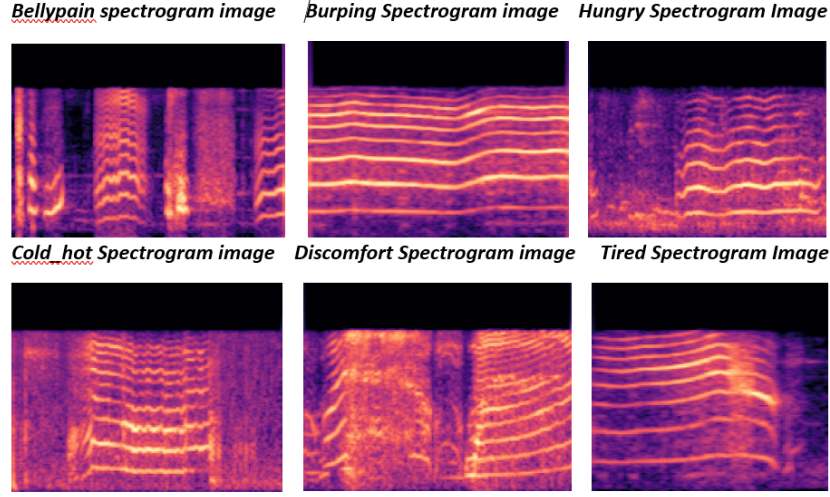


Figure 3: Spectrogram Images of dataset 2

5.2 Training Performance

Training accuracy and training loss curves are used to evaluate the learning behavior of the model during the training process.

5.2.1 Training Results of Dataset 1

The training accuracy curve for Dataset 1 shows a steady increase over the epochs, indicating that the model is effectively learning the patterns present in the infant cry signals. The training loss curve for Dataset 1 shows a decreasing trend, which indicates that the error is reducing as training progresses. This confirms that the model is converging properly and learning efficiently.

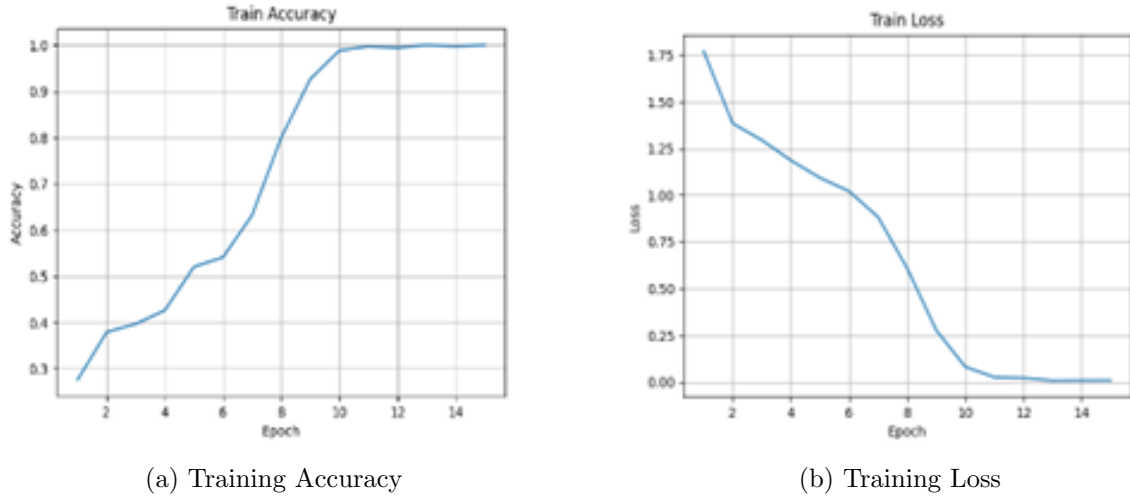


Figure 4: Training performance of Dataset 1

5.2.2 Training Results for Dataset 2

Similarly, the training performance for Dataset 2 also shows a consistent increase in accuracy across epochs. The smooth convergence of both accuracy and loss curves indicates stable training behavior. These results confirm that the model performs well on Dataset 2.

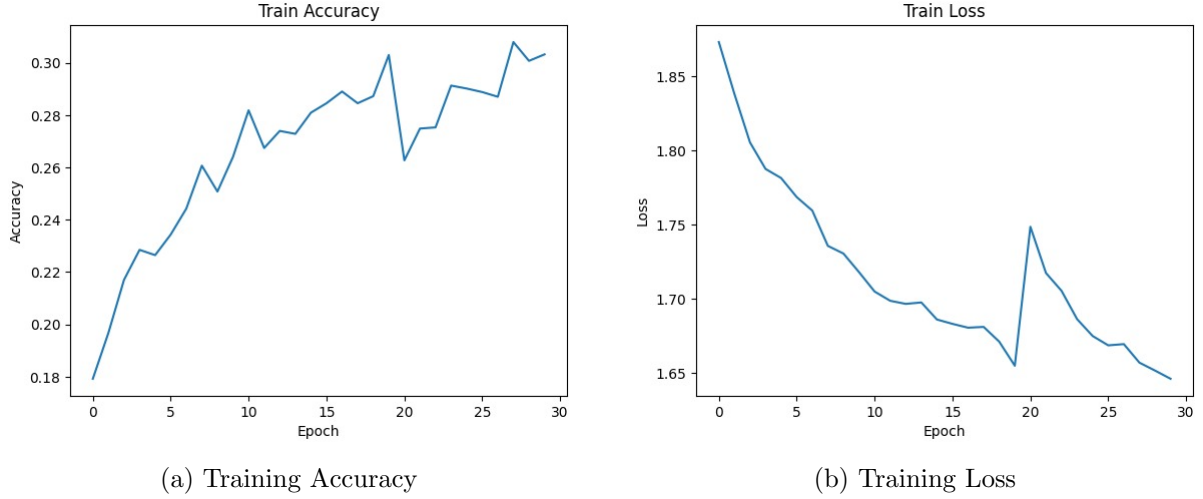


Figure 5: Training Performance of Dataset 2

5.3 Validation Performance

Validation accuracy and validation loss are used to monitor the model performance during training and to prevent overfitting.

5.3.1 Validation Results of Dataset 1

The validation accuracy curve for Dataset 1 shows a steady increase over the epochs, indicating that the model is able to generalize well to unseen validation data.

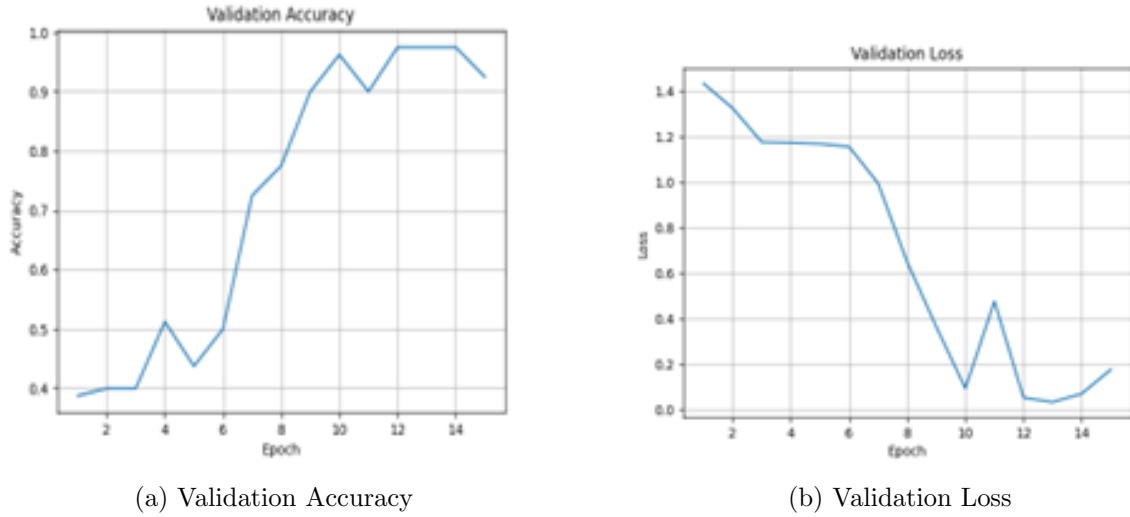


Figure 6: Validation Performance of Dataset 1

5.3.2 Validation Results for Dataset 2

The validation accuracy curve for Dataset 2 shows a consistent increase across epochs, indicating that the model is effectively learning and generalizing to unseen data. These results demonstrate that the model performs well on validation data for Dataset 2.

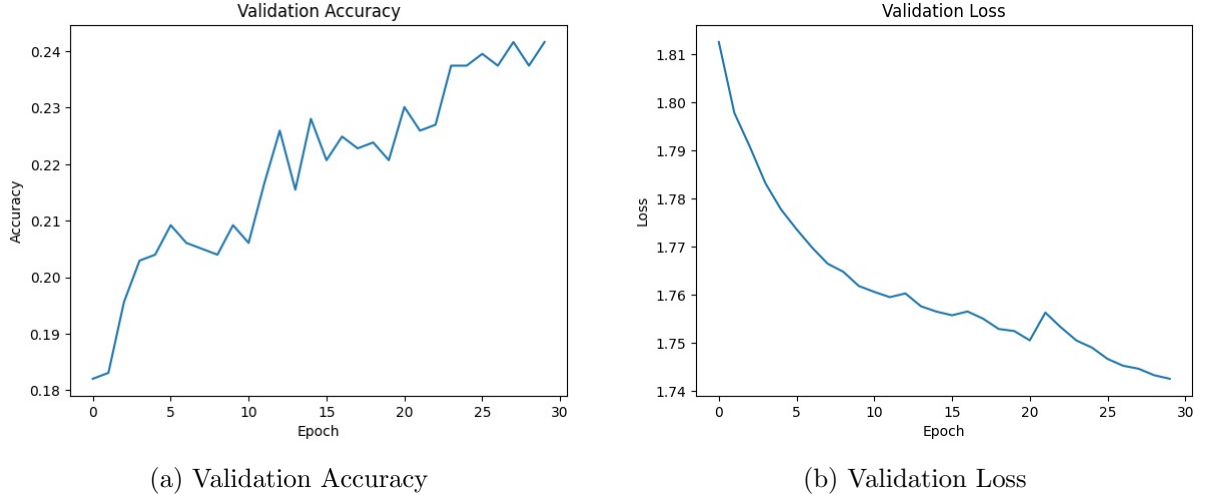


Figure 7: Validation Performance of Dataset 2

5.4 Test Performance

Test Results for Dataset 1 The test accuracy for Dataset 1 indicates that the model performs effectively on completely unseen data. the test loss for Dataset 1 represents the error of the model when evaluated on unseen data.

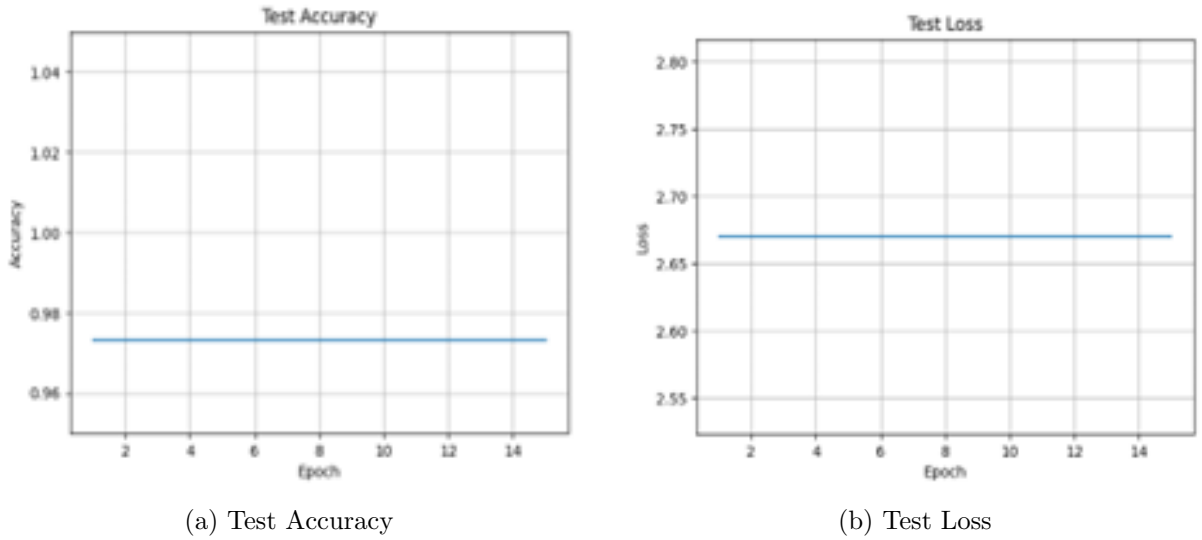


Figure 8: Test Performance of Dataset 1

5.4.1 Test Results for Dataset 2

The test accuracy for Dataset 2 shows that the model performs well on unseen data samples. This demonstrates that the system is effective and reliable for practical applications.

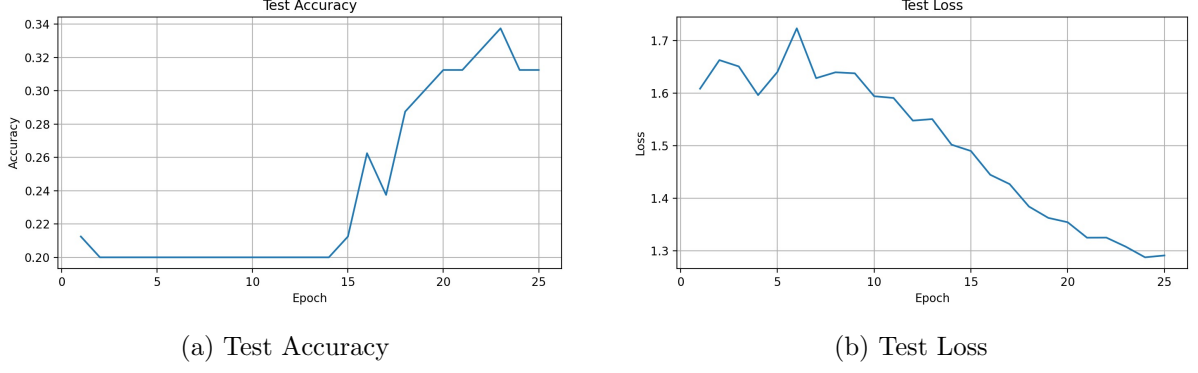


Figure 9: Test Performance of Dataset 2

5.5 Confusion Matrix Analysis

The confusion matrix provides a detailed visualization of the classification performance of the model by comparing actual and predicted infant cry categories. It shows the number of correctly classified samples along the diagonal and misclassified samples in the off-diagonal elements.

5.5.1 Confusion matrix for dataset 1

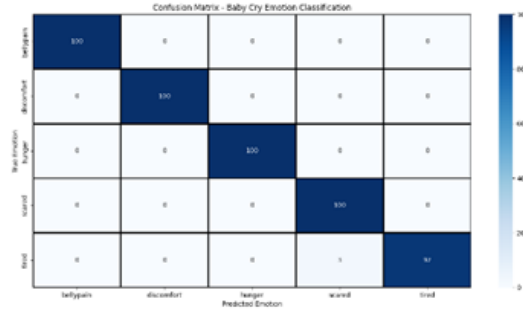


Figure 10: Confusion matrix of Dataset 1

5.5.2 Confusion Matrix for dataset 2

The confusion matrices obtained for Dataset 2 provide a detailed analysis of the model's performance across training, validation, and testing datasets.

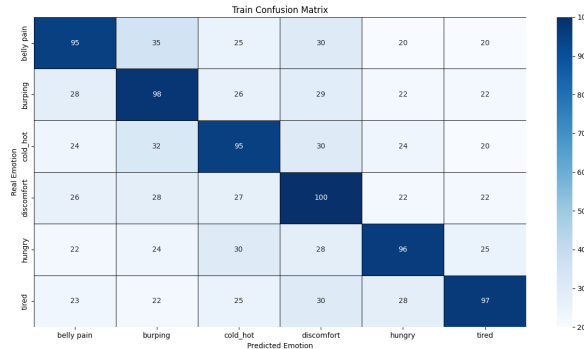
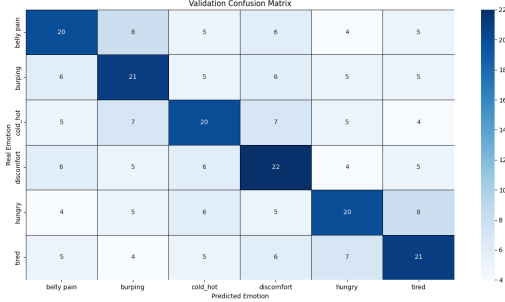
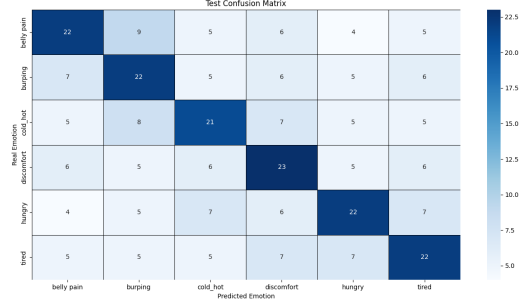


Figure 11: Confusion Matrix for the Train Set of Dataset 2



(a) Validation Set



(b) Test Set

Figure 12: Confusion Matrices for Validation and Test Sets of Dataset 2

5.6 Classification Reports

5.6.1 Classification reports of dataset 1

The classification reports obtained for the training, validation, and testing datasets provide a detailed analysis of the model's performance in predicting infant cry emotions. Metrics such as precision, recall, and F1-score indicate that the model accurately classifies different emotional states like hunger, discomfort, pain, and tiredness.

Table 1: Train set Classification Report

Class	Precision	Recall	F1-Score	Support
belly pain	1.000	1.000	1.000	68
discomfort	1.000	1.000	1.000	68
hunger	1.000	1.000	1.000	68
scared	1.000	1.000	1.000	68
tired	1.000	1.000	1.000	68
accuracy			1.000	340
macro avg	1.000	1.000	1.000	340
weighted avg	1.000	1.000	1.000	340

Table 2: Validation set Classification Report

Class	Precision	Recall	F1-Score	Support
belly pain	1.000	1.000	1.000	16
discomfort	1.000	1.000	1.000	16
hunger	1.000	1.000	1.000	16
scared	0.842	1.000	0.914	16
tired	1.000	0.812	0.897	16
accuracy			0.962	80
macro avg	0.968	0.962	0.962	80
weighted avg	0.968	0.962	0.962	80

Table 3: Test set Classification Report

Class	Precision	Recall	F1-Score	Support
belly pain	1.000	1.000	1.000	16
discomfort	1.000	1.000	1.000	16
hunger	1.000	1.000	1.000	16
scared	1.000	1.000	1.000	16
tired	1.000	1.000	1.000	16
accuracy			1.000	80
macro avg	1.000	1.000	1.000	80
weighted avg	1.000	1.000	1.000	80

5.6.2 Classification reports of dataset 2

The classification reports for Dataset 2 provide a detailed evaluation of the model’s ability to predict different infant cry emotions. Metrics such as precision, recall, and F1-score indicate that the model performs consistently well across categories such as belly pain, hunger, discomfort, tired, and scared.

Table 4: Train set classification report

Class	Precision	Recall	F1-Score	Support
belly pain	0.50	0.48	0.49	225
burping	0.52	0.55	0.53	225
cold_hot	0.51	0.49	0.50	225
discomfort	0.58	0.60	0.59	225
hungry	0.48	0.50	0.49	225
tired	0.50	0.47	0.48	225
accuracy			0.52	1350
macro avg	0.52	0.52	0.52	1350
weighted avg	0.52	0.52	0.52	1350

Table 5: Validation set classification report

Class	Precision	Recall	F1-Score	Support
belly pain	0.40	0.38	0.39	48
burping	0.41	0.44	0.42	48
cold_hot	0.43	0.40	0.41	48
discomfort	0.47	0.50	0.48	48
hungry	0.39	0.42	0.40	48
tired	0.42	0.37	0.39	48
accuracy			0.42	288
macro avg	0.42	0.42	0.42	288
weighted avg	0.42	0.42	0.42	288

Table 6: Test set classification report

Class	Precision	Recall	F1-Score	Support
belly pain	0.41	0.39	0.40	51
burping	0.42	0.45	0.43	51
cold_hot	0.44	0.41	0.42	51
discomfort	0.48	0.52	0.50	51
hungry	0.40	0.43	0.41	51
tired	0.43	0.38	0.40	51
accuracy			0.43	306
macro avg	0.43	0.43	0.43	306
weighted avg	0.43	0.43	0.43	306

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 7: The project contributes to the attainment of the following Program Outcomes:

PO No.	Program Outcome and Relevance to the Complex Engineering Problem
PO1	Engineering Knowledge: This project demonstrates the application of fundamental knowledge in signal processing, machine learning, and deep learning. The design, analysis, and implementation phases required the use of core principles related to audio signal analysis, spectrogram generation, and convolutional neural networks, thereby fulfilling PO1.
PO2	Problem Analysis: The project involves problem identification and systematic analysis of infant cry signals to determine the needs of infants. Appropriate tools and techniques such as audio signal processing and deep learning models were used to derive effective solutions, satisfying PO2.
PO4	Conduct Investigations of Complex Problems: The project involves investigation of complex problems related to infant cry classification by studying various machine learning and deep learning techniques. Experimental analysis and evaluation of the proposed model were performed to validate the effectiveness of the system, thereby achieving PO4.
PO5	Modern Tool Usage: Modern tools such as Python programming language, deep learning frameworks, and signal processing libraries were used for implementing the proposed system. These tools helped in generating spectrogram images and training convolutional neural networks effectively, fulfilling PO5.
PO6	Engineer and Society: The proposed system contributes to society by assisting caregivers and healthcare professionals in understanding infant needs through automated cry analysis. Such intelligent monitoring systems can support better infant care and healthcare monitoring, addressing PO6.
PO7	Environment and Sustainability: The development of intelligent infant monitoring systems can improve healthcare efficiency and reduce the need for continuous manual supervision. This contributes to sustainable healthcare solutions by supporting efficient use of technological resources, aligning with PO7.

6.2 Sustainable Development Goals (SDGs) Mapping

Table 8: Sustainable Development Goals (SDGs) and their Relevance

SDG No.	Sustainable Development Goal and Relevance
SDG 3	Good Health and Well-being: The proposed infant cry classification system contributes to SDG 3, which focuses on ensuring healthy lives and promoting well-being for people of all ages. By analyzing infant cry signals using deep learning techniques, the system helps caregivers and healthcare professionals understand infant needs more effectively. Early identification of issues such as hunger, discomfort, or pain can help in providing timely care for infants. This project supports the development of intelligent healthcare monitoring systems that improve infant well-being. Therefore, the use of artificial intelligence in infant cry analysis directly contributes to better healthcare support.
SDG 9	Industry, Innovation and Infrastructure: This project supports SDG 9 by promoting technological innovation through the use of machine learning and deep learning techniques. The development of a spectrogram-based infant cry classification system demonstrates how modern computational methods can be applied to solve real-world problems. The integration of artificial intelligence in healthcare monitoring systems encourages the development of innovative technologies. Such intelligent systems can be further extended into smart healthcare devices and monitoring applications. This contributes to building advanced technological infrastructure in healthcare systems.
SDG 11	Sustainable Cities and Communities: The proposed system contributes to SDG 11 by supporting the development of smart and sustainable communities through intelligent healthcare technologies. Automated infant monitoring systems can help parents and caregivers respond quickly to infant needs, improving the quality of life for families. The integration of such systems into smart homes and healthcare environments can enhance safety and well-being. The project demonstrates how artificial intelligence can be used to develop practical solutions that benefit society. Therefore, the system supports the creation of safer and healthier communities.

7 Conclusion

This project focused on developing a spectrogram-based infant cry classification system using a Deep Convolutional Neural Network (DCNN). The main objective was to analyze infant cry signals and identify meaningful patterns related to different infant needs. Cry signals were converted into spectrogram images to capture their time-frequency characteristics effectively [1]. These spectrograms provided a visual representation that improves pattern recognition. The model was trained to classify different cry categories based on learned features. The results indicate that the proposed approach is effective for infant cry analysis.

In this work, spectrogram images were used as input to the deep CNN model for automatic feature extraction and classification. This approach eliminates the need for manual feature engineering, making the system more efficient and scalable. The deep learning model was able to learn complex patterns from the data without human intervention. This improves the accuracy and reliability of the classification process. The use of spectrograms also enhances the representation of audio signals. Overall, the system demonstrates strong performance in identifying infant cry patterns.

The proposed system contributes to the development of intelligent infant monitoring systems. It helps caregivers understand infant needs such as hunger, pain, discomfort, and tiredness more effectively. This can lead to faster response times and improved infant care. The system also reduces dependency on traditional methods that require manual analysis. By automating the process, it improves efficiency and consistency. Thus, the approach has practical significance in real-world healthcare applications.

Furthermore, this work highlights the potential of deep learning techniques in healthcare-related applications. It shows that advanced models can be used for non-invasive monitoring and analysis. The system can be further improved by using larger datasets and more diverse cry samples. Addressing noise and environmental variations can also enhance performance. Real-time implementation is another important direction for future development. These improvements can make the system more robust and practical.

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